Observation Project: The Constellation Hunt

Practical project — at least 3 observing sessions 3–4 weeks

Name: _____ Date: _____ Marks: _____ / 60

Goal

Become someone who can walk outside, look up and **name what they see**: by the end, find at least **8 constellations or asterisms and 10 bright stars** without help.

Before you start: reread Lecture 2.5. Take a seasonal chart, a red torch, this sheet and a pencil. Dark-adapt for 20 minutes. Go with a friend or family member.

Task 1 — Sky conditions log

6 marks

#	Date	Time (IST)	Place	Moon / tithi	Clouds (0–8)	Little-Dipper stars seen
1						
2						
3						

Task 2 — Find, sketch and confirm

24 marks

Find **six** constellations/asterisms, including at least two signposts of the current season. For each: sketch it **as you see it** (mark the horizon side and draw brighter stars larger); record its direction and altitude (fists); describe the star-hop you used; confirm with an app or chart *after* sketching. (4 marks each: identification, sketch, direction/altitude, star-hop.)

#	Name	Time	Direction	Altitude	How I found it (star-hop)	✔
1						
2						
3						
4						
5						
6						

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1. horizon ↓	2. horizon ↓	3. horizon ↓
4. horizon ↓	5. horizon ↓	6. horizon ↓

Task 3 — Star colours

8 marks

Find **four** first-magnitude stars visible this season and compare their colours by eye (slightly defocused binoculars help).

Star	Colour I see	Colour from Lecture 2.4	Approx. temperature

Which looked the reddest, and which the bluest? Did your eyes agree with the table?

Task 4 — Measuring the sky

12 marks

(a) Latitude from Polaris (4). Measure Polaris's altitude three times (fist method or phone clinometer); average; compare with your real latitude. If Polaris is hidden by haze or buildings (common in south India), do the noon shadow method of Assignment 1C instead.

Try 1	Try 2	Try 3	Average	Real latitude	Difference

(b) Angular distances (4). With your hand, measure the length of one asterism (e.g. Orion's Belt or the Big Dipper) and the distance between two bright stars (e.g. Castor–Pollux, Vega–Altair, Arcturus–Spica). Compare with the true values from an app.

(c) The sky turns (4). Note a bright star's position; measure again exactly 1 hour later. How many degrees did it move? (Expected $\approx 15^\circ$ along its daily circle.)

Task 5 — Reflection **10 marks**

In 150–200 words: Which constellation was hardest to find, and why? How did light pollution, the Moon or clouds affect what you saw? Did anything surprise you (a meteor, a satellite, a planet, a star's colour)? Which star-hop will you teach to someone else?

Bonus (up to 5): teach one constellation to a family member or friend and get them to point it out to you — write their name and what they said!

Task	1	2	3	4	5	**Total**
Marks	6	24	8	12	10	**60**